

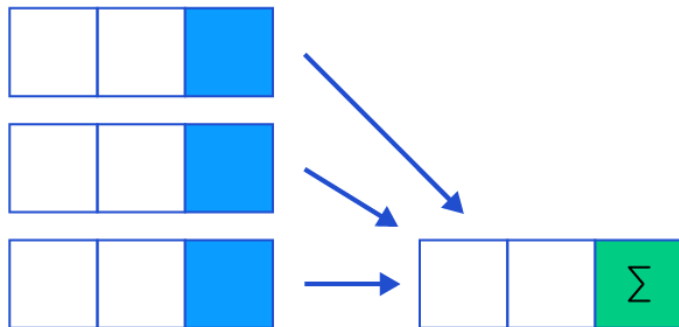
SQL Optimization and Advanced Querying

What is Window Functions?

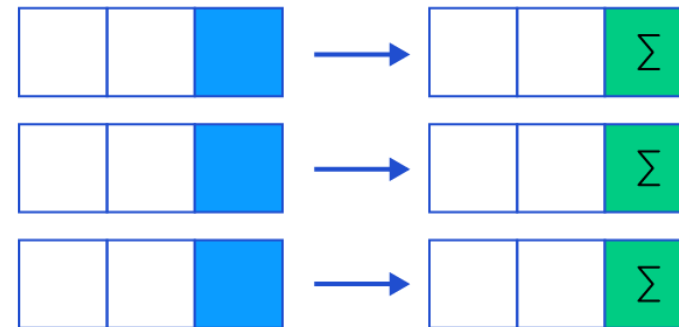
Window functions are used to **perform a calculation on an aggregate value** based on a set of rows.

Return multiple rows for each group.

Aggregate Functions (SUM, AVG, etc.)



Window Functions (Over, Partition, Order, etc.)



Window Functions: Syntax

```
window_function_name([ALL] expression)
OVER (
  [PARTITION BY ...]
  [ORDER BY ...]
)
```

ALL (optional): count all values along with duplicates.

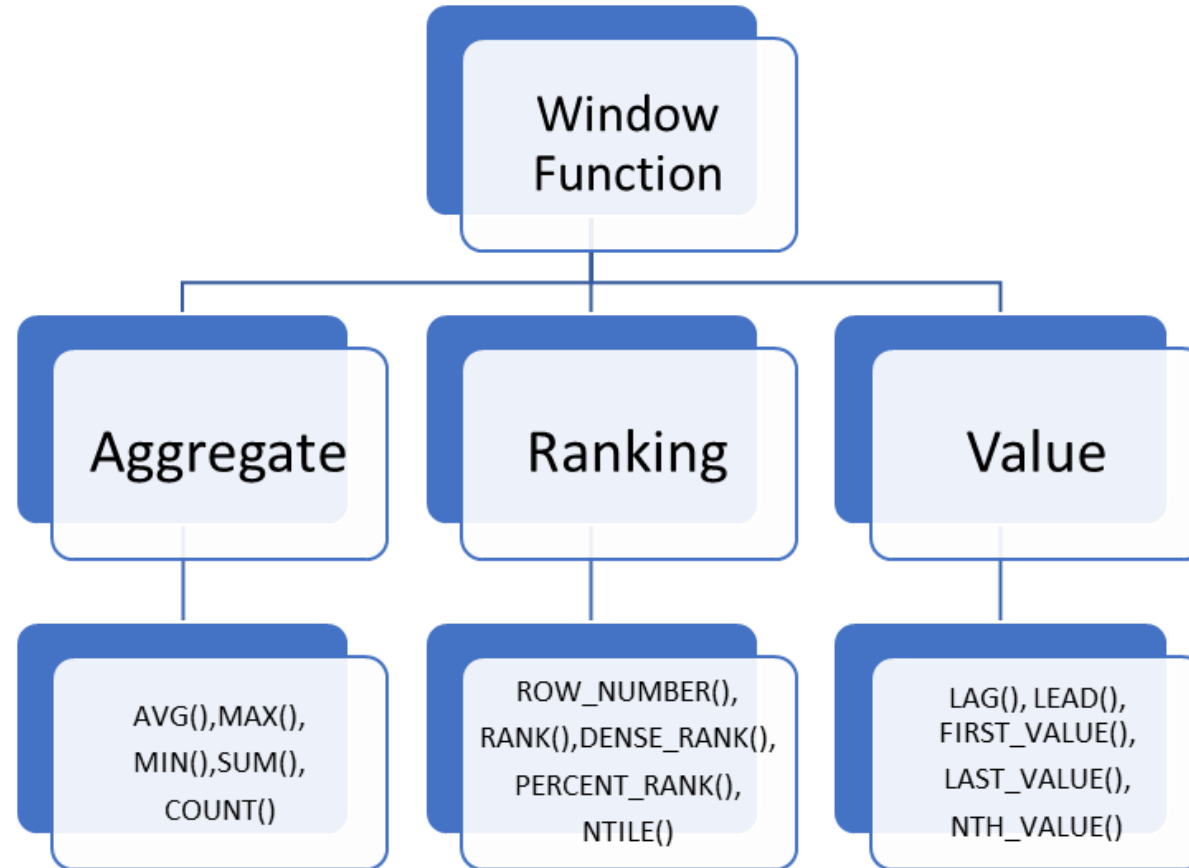
We cannot use the DISTINCT keyword in window functions.

Expression: the column name for which we calculate an aggregated value.

Window Functions: Clauses

- **OVER:** defines the window.
- **PARTITION BY:** splits rows into partitions for independent calculations.
- **ORDER BY:** defines row order within each partition.

Types of Window Functions



Example: Sales Table

```
CREATE TABLE Sales (  
  Employee VARCHAR(45) NOT NULL,  
  Year INT NOT NULL,  
  Country VARCHAR(45) NOT NULL,  
  Product VARCHAR(45) NOT NULL,  
  Amount DECIMAL(12,2) NOT NULL,  
  PRIMARY KEY (Employee, Year)  
);
```

Aggregate Window Functions - SUM

```
SELECT Employee, Year, Country,  
        Product, Amount,  
        SUM(Amount) OVER (PARTITION BY Country)  
AS Total  
FROM Sales;
```

SUM Output

Employee	Year	Country	Product	Amount	Total
Or Peretz	2017	Israel	Computer	15000	45000
Or Peretz	2018	Israel	Computer	10000	45000
Or Peretz	2019	Israel	TV	20000	45000
Omer Doron	2018	USA	TV	20000	30000
Omer Doron	2019	USA	Mobile	10000	30000

Aggregate Window Functions - AVG

```
SELECT Employee, Year,  
        Country, Product, Amount,  
        AVG(Amount) OVER (PARTITION BY Country,  
YEAR(Year)) AS AvgSales  
FROM Sales;
```

AVG Output

Employee	Year	Country	Product	Amount	AvgSales
Or Peretz	2017	Israel	Computer	15000	15000
Or Peretz	2018	Israel	Computer	10000	15000
Or Peretz	2019	Israel	TV	20000	15000
Omer Doron	2018	USA	TV	20000	15000
Omer Doron	2019	USA	Mobile	10000	15000

Aggregate Window Functions - COUNT

```
SELECT Employee, Year,  
        Country, Product, Amount,  
        COUNT(Product) OVER (PARTITION BY  
Country) AS TotalProduct  
FROM Sales;
```

Aggregate Window Functions - MAX

```
SELECT Employee, Year,  
        Country, Product, Amount,  
        MAX(Product) OVER (PARTITION BY Country)  
AS TotalProduct  
FROM Sales;
```

Ranking Window Functions - Data for Examples

FirstName	LastName	City
Luisa	Evans	Texas
Paul	Ward	Alaska
Peter	Bennett	California
Carlos	Patterson	New York
Rose	Huges	Florida
Marielia	Simmons	Texas
Antonio	Butler	New York
Diego	Cox	California

Example - RANK

```
SELECT FirstName, LastName, City,  
        RANK() OVER (ORDER BY City) AS RankNo  
FROM table;
```

RANK Output

FirstName	LastName	City	RankNo
Paul	Ward	Alaska	1
Peter	Bennett	California	2
Diego	Cox	California	2
Rose	Huges	Florida	4
Carlos	Patterson	New York	5
Antonio	Butler	New York	5
Luisa	Evans	Texas	7
Marielia	Simmons	Texas	7

Example - DENSE_RANK

```
SELECT FirstName, LastName, City,  
        DENSE_RANK() OVER (ORDER BY City) AS  
RankNo  
FROM table;
```


Example - NTILE

```
SELECT FirstName, LastName, City,  
        NTILE(3) OVER (ORDER BY City) AS RankNo  
FROM table;
```

Value Window Functions - LEAD

```
SELECT Year, Product, Country, Amount,  
        LEAD(Amount, 1) OVER (PARTITION BY Year  
ORDER BY Country) AS NextAmount  
FROM Sales;
```

Value Window Functions - LAG

```
SELECT Year, Product, Country, Amount,  
        LAG(Amount, 1) OVER (PARTITION BY Year  
ORDER BY Country) AS PrevAmount  
FROM Sales;
```

FIRST_VALUE / LAST_VALUE

```
SELECT Year, Product, Country, Amount,  
        FIRST_VALUE(Amount) OVER (PARTITION BY  
Country ORDER BY Country) AS FirstAmount,  
        LAST_VALUE(Amount) OVER (PARTITION BY  
Country ORDER BY Country) AS LastAmount  
FROM Sales;
```

Key Takeaways

- Correct join choice controls both correctness and NULL behavior.
- Subqueries with `ALL`, `IN`, and `EXISTS` express different logical checks.
- Window functions enable analytics patterns without losing row-level detail.